Lab: Redshift Process Layer Ingestion using AWS Glue (Star Schema Load)

Summary of lab:

In this lab, we deploy a Glue Python Shell job that ingests data from the **Redshift Raw Layer** into the **Process Layer (Curated Layer)**.

The job performs:

- Staging using **temporary tables**
- Merging data into:
  - `dim_apartments`
  - `dim_users`
  - `fact_apartment_viewings`
- Incremental loading for the fact table using the maximum processed timestamp

After successful execution, the **Process Zone** in Redshift contains fully merged dimensional and fact tables following a star schema design.

Prerequisites:

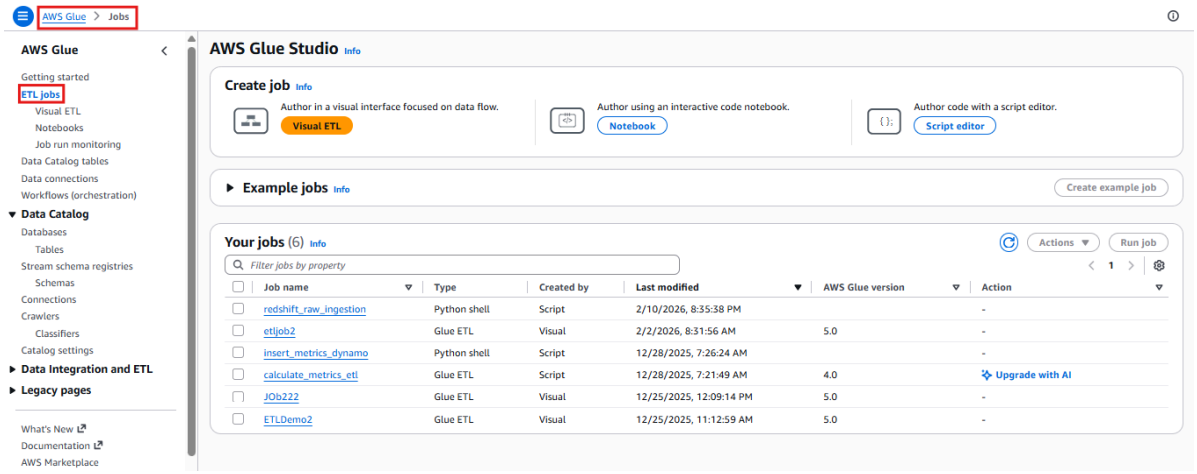
- Redshift Serverless namespace and workgroup created
- Database created: `db_rental_apartments`
- Schemas created:
  - `raw_zone`
  - `process_zone`
- Raw layer tables populated:
  - `raw_zone.apartments`

    - `raw_zone.apartment_attributes`
    - `raw_zone.apartment_viewings`
- Process layer tables created:
    - `process_zone.dim_apartments`
    - `process_zone.dim_users`
    - `process_zone.fact_apartment_viewings`
- AWS Secrets Manager secret configured for Redshift credentials
- IAM role for Glue with access to:
    - Redshift
    - Secrets Manager
    - S3
    - CloudWatch Logs
- Script file: `redshift_process_layer.py`

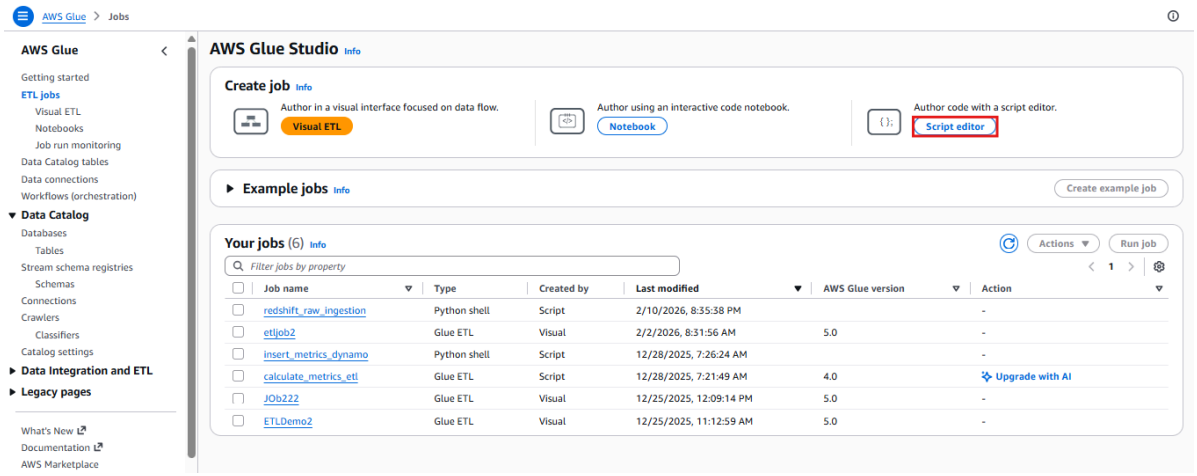
Steps to complete lab:

1. Create Glue Job

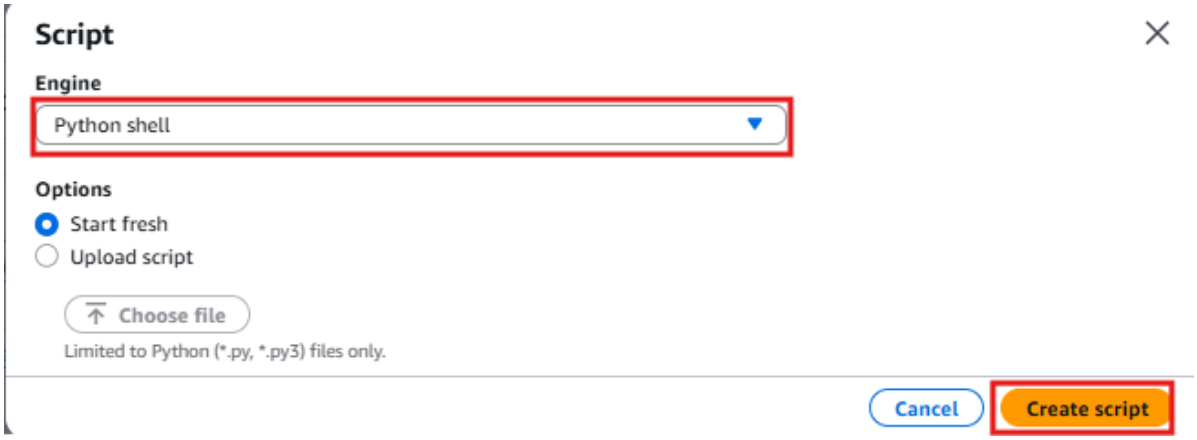
- Navigate to: AWS Console → Glue → ETL Jobs



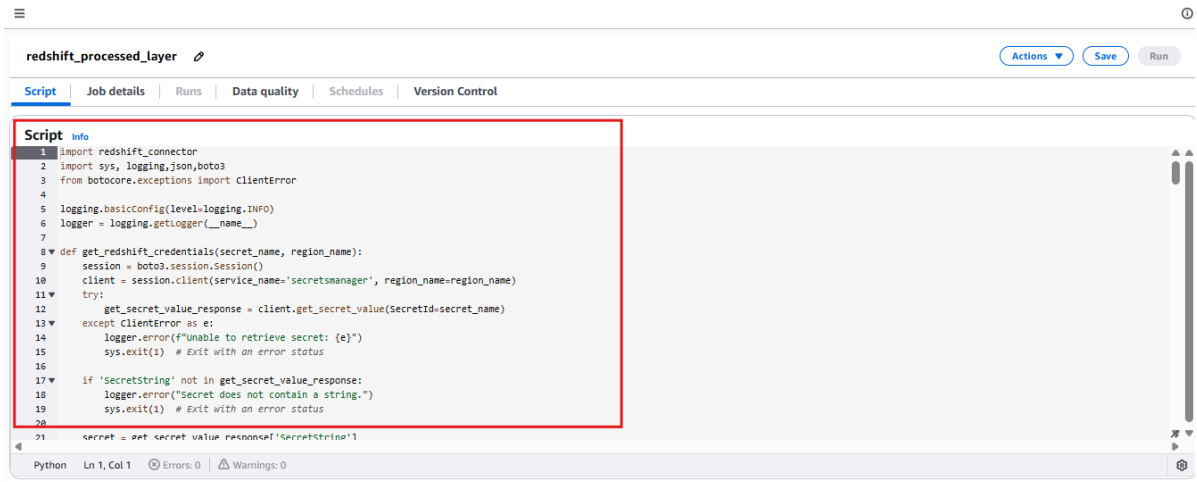
- Click **Script Editor**



- Select **Python Shell**

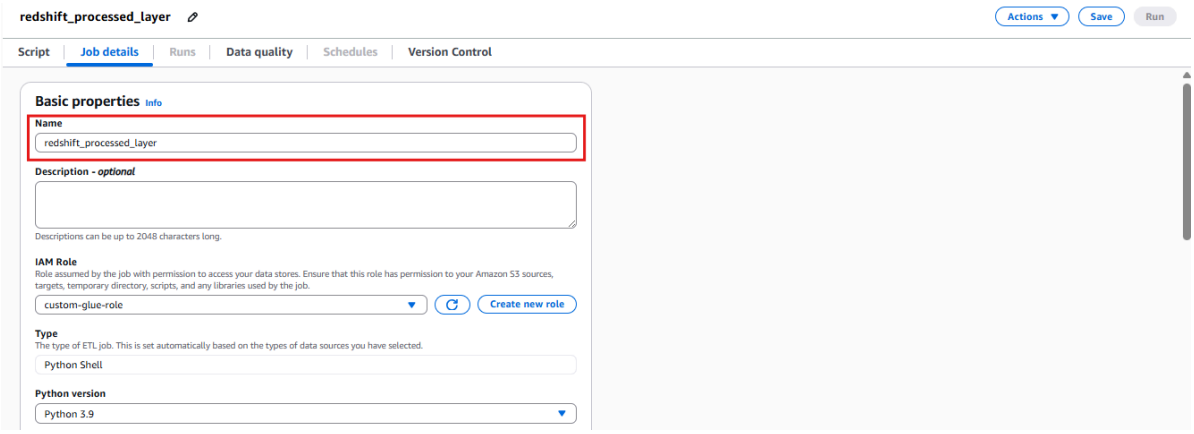


- Paste the script contents

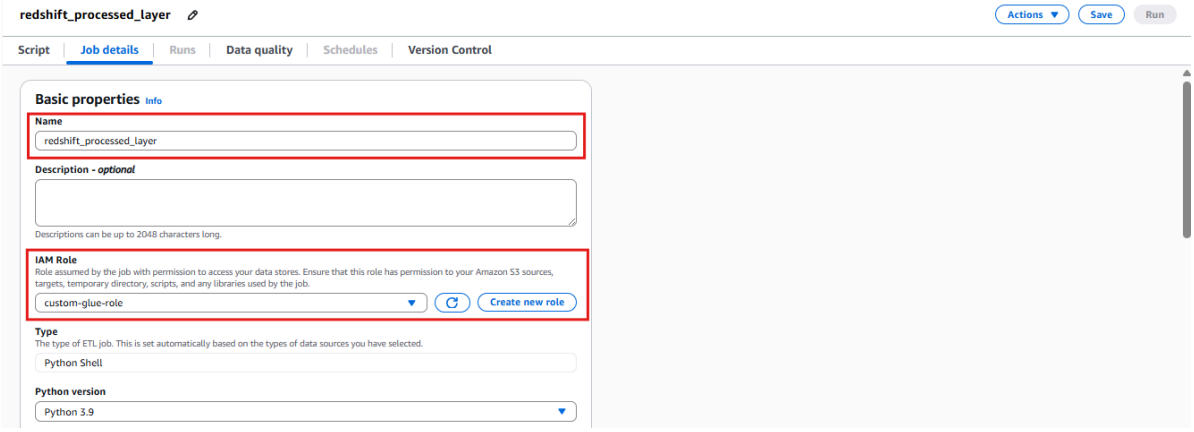


2. Configure Job Details

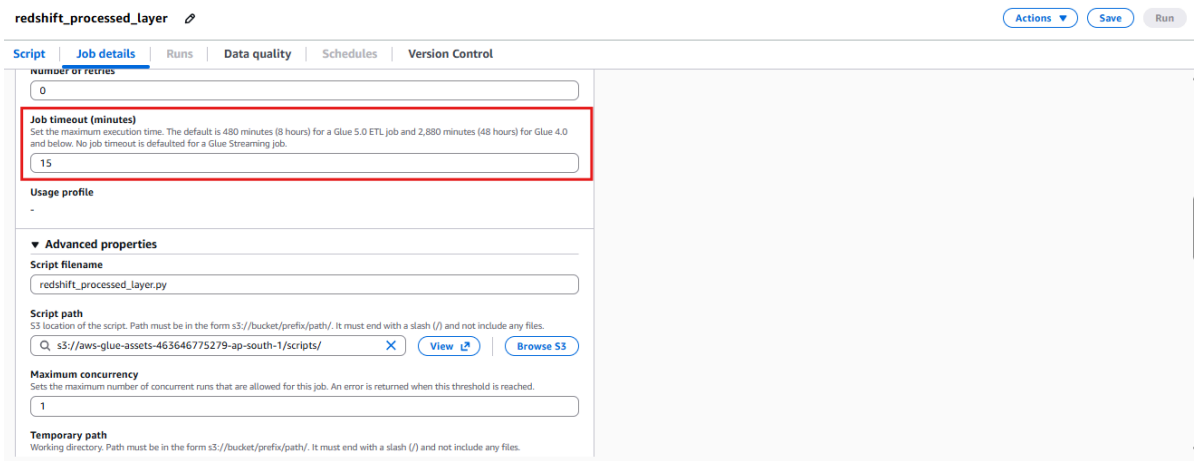
- Job Name: `Redshift_Process_Layer`



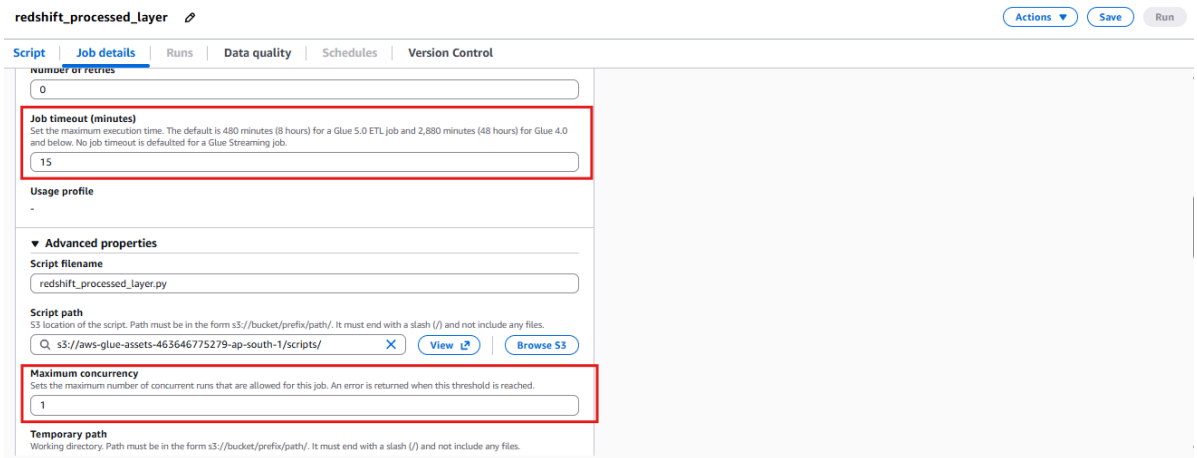
- IAM Role: Select custom Glue role



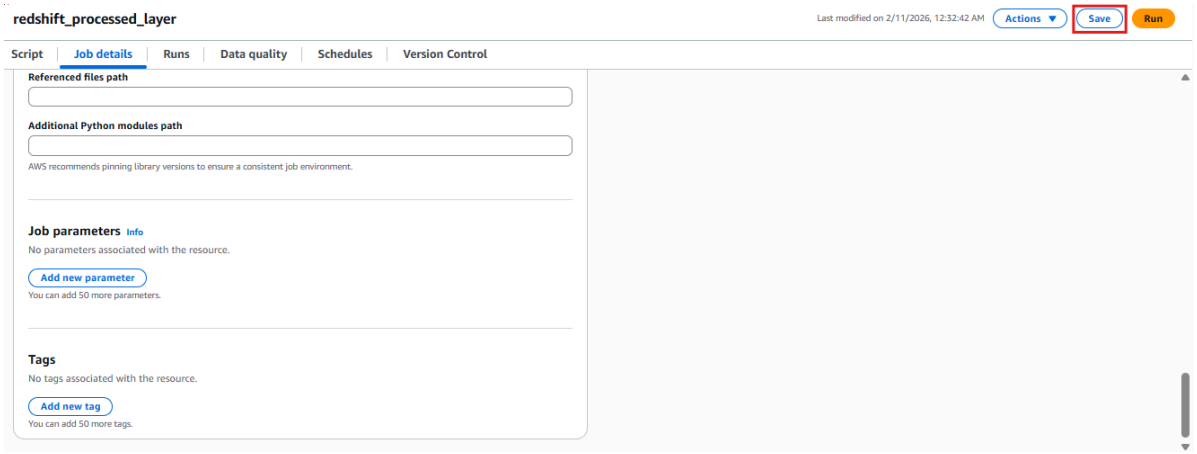
- Timeout: 15 minutes



- Max Concurrency: 1

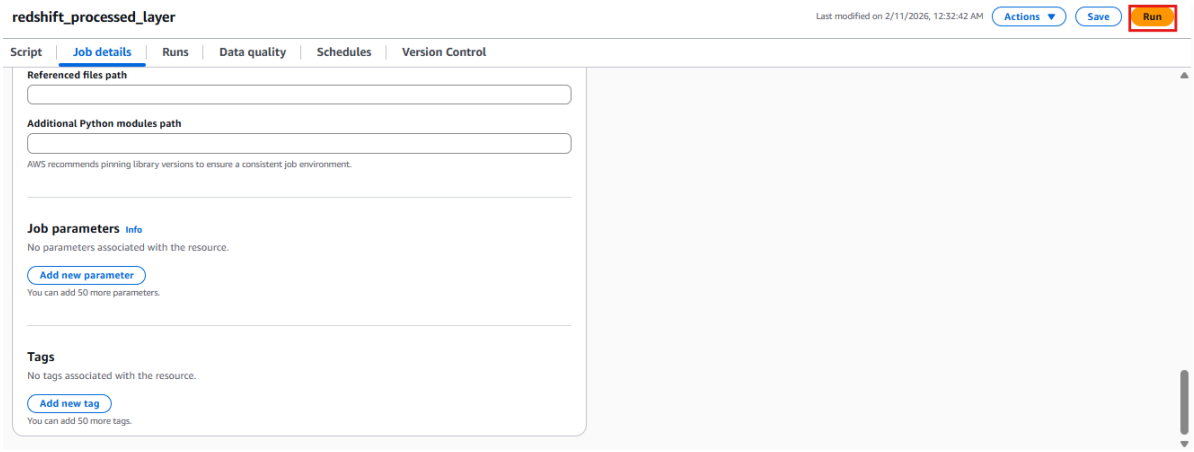


- Click **Save**

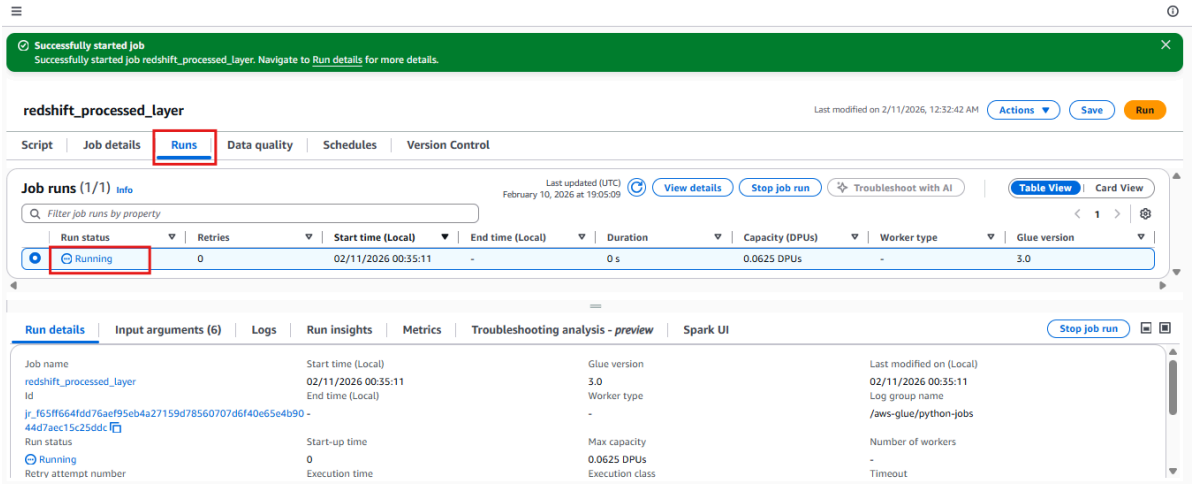


3. Run the Job

- Click **Run**



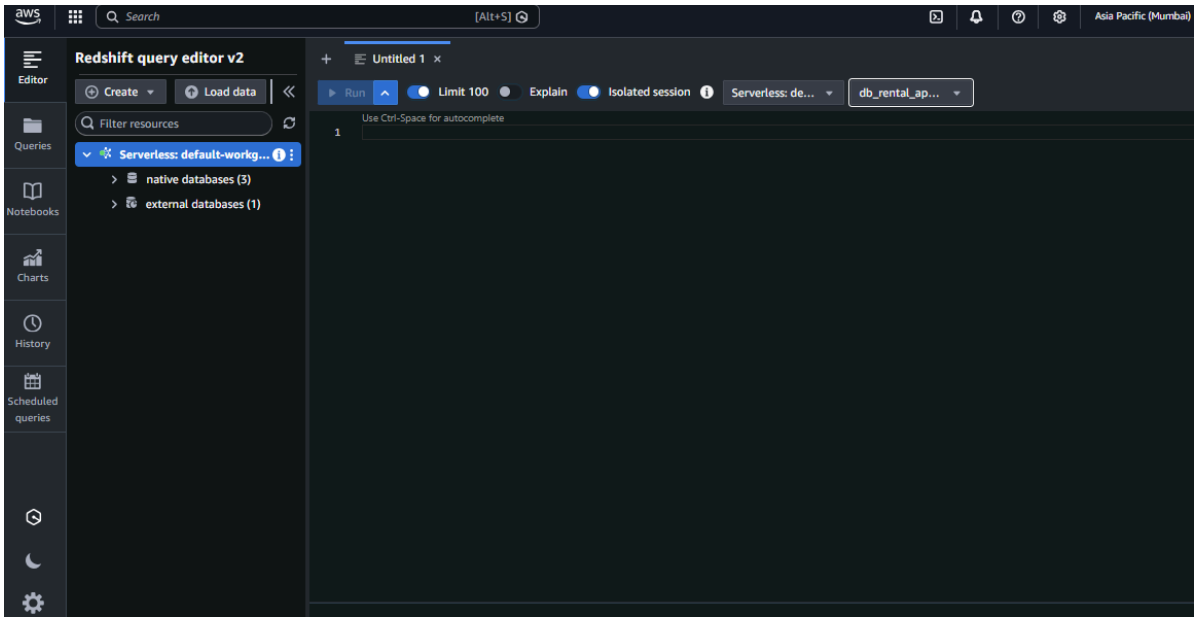
- Monitor execution in the **Runs** tab



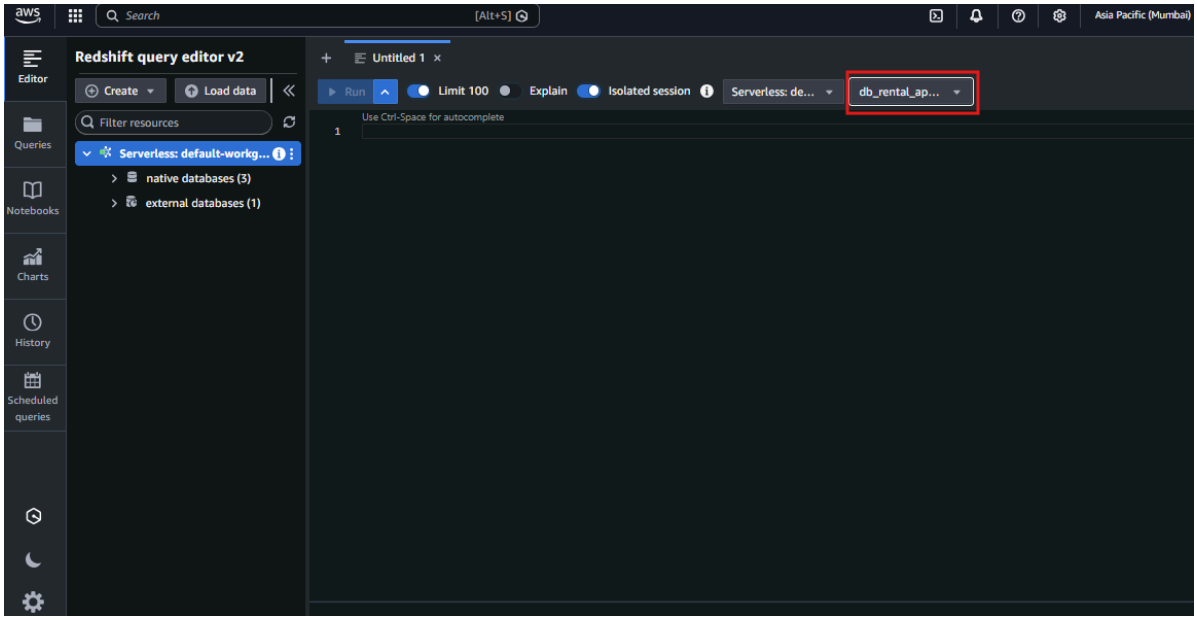
- Wait until status shows **Succeeded**

4. Verify Data in Redshift

- Go to Redshift → Workgroup → Query Data



- Select database: `db_rental_apartments`



Run:

```
select * from process_zone.dim_apartments limit 10;
```

```
select * from process_zone.dim_users limit 10;
```

```
select * from process_zone.fact_apartment_viewings limit 10;
```

Verification/outcome:

- Glue job completes successfully
- Temporary staging tables are created and dropped automatically
- `dim_apartments` populated with joined apartment and attribute data
- `dim_users` populated
- `fact_apartment_viewings` loaded incrementally
- Raw layer → Process layer transformation successful